

Bachelor Thesis Exposé

Autonomous Mobile Manipulation: Design and Evaluation of a Pick-and-Place and Object Stacking Pipeline on the LIMO Cobot

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Title

Primary Title:

Autonomous Mobile Manipulation: Design and Evaluation of a Pick-and-Place and Object Stacking Pipeline on the LIMO Cobot

Alternative titles:

- Autonomous Mobile Manipulation on the LIMO Cobot: Design, Evaluation, and Simulation-to-Reality Transfer of a Pick-and-Place Pipeline
- Autonomous Mobile Manipulation: Design and Evaluation of a Pick-and-Place Pipeline on a Table-Sized Mobile Manipulator (LIMO Cobot)

Research Question

How can a modular autonomous mobile-manipulation pipeline, integrating onboard navigation, perception, and collision-aware motion planning, be designed and evaluated in a ROS2/Gazebo simulation to perform reliable, repeatable pick-and-place and object-stacking tasks with a compact manipulator and gripper, and to what extent can it then be transferred to and validated on the physical robot?

Sub-Questions

1. **Navigation.** How reliably and repeatably can an autonomous mobile base position itself such that a target object lies within the manipulator's reachable workspace, and to what

extent does base-positioning accuracy determine subsequent grasp success?

- 2. Perception.** How accurately and robustly can a target object's three-dimensional pose be estimated from onboard RGB-D sensing and expressed in the manipulator's planning frame, and how does the resulting estimation error propagate to grasp success?
- 3. Manipulation and reachability.** To what extent do the arm-mounting configuration, base stop distance, and workspace geometry constrain the reachable workspace of a short-reach manipulator, and how can collision-aware motion planning ensure that grasp and placement trajectories remain safe and executable within those constraints?
- 4. Stacking precision.** With what precision and repeatability can objects be placed to achieve stable sequential vertical stacking, and which sources of error most strongly limit stacking reliability?
- 5. Evaluation in simulation.** What level of task performance (success rate, placement accuracy, planning time, and cycle time) does the integrated pipeline achieve under repeated trials, and which failure modes dominate?
- 6. Simulation-to-reality transfer.** To what extent does a pipeline developed in simulation transfer to the physical robot, how does real-world performance compare with simulated performance, and what engineering effort does the transfer entail?

Scope and Limitations

Scope (Delimitations)

- ✓ The thesis designs and evaluates a complete mobile-manipulation pipeline covering navigation, perception, grasping, placement, and sequential stacking, for a single compact mobile manipulator (AgileX LIMO Cobot).
- ✓ The system is developed and primarily evaluated in physics-based simulation; deployment on the physical robot is pursued as a transfer-and-validation objective, contingent on hardware availability and the time available within the thesis period.
- ✓ The operating environment is *static, planar, and structured*: a known layout, known support surfaces, and a single known target object at a time.
- ✓ Perception is delimited to a single, geometrically simple, colour-distinct object resting on a known support surface.
- ✓ Manipulation is restricted to top-down grasps of light, regularly shaped objects lying within the arm’s reachable workspace.
- ✓ Evaluation is quantitative, based on success rate, placement accuracy, planning time, and cycle time, assessed under repeated trials ($N \geq 20$).

Limitations

- Results are established primarily in simulation; despite physics-based modelling, a simulation-to-reality gap may affect real-world performance, and characterising this gap is itself part of the study.
- The limited reach of the manipulator (≈ 0.28 m) constrains the feasible workspace and the admissible placement of objects and support surfaces.
- Classical colour-and-depth perception is sensitive to lighting, sensor noise, and object appearance; robustness under uncontrolled conditions is not claimed.
- Achievable stacking precision is bounded by the combined error of base positioning, perception, and end-effector alignment.
- Physical-hardware validation depends on platform availability, calibration, and the time remaining within the thesis period.

Out of Scope

- Dynamic or deformable environments and moving obstacles/targets.
- Cluttered or multi-object scenes and general object recognition.
- Learning-based perception and learned manipulation policies.
- Multi-robot coordination.
- Online map-building (SLAM) during execution; a known map and layout are assumed.

Task Planning

Total thesis period: 3 months (13 weeks). Phases overlap deliberately to make the most of the available time. Continuous light drafting runs throughout Phases 1–4 so that the writing phase is consolidation rather than writing from scratch.

Phase 1 — Consolidation (Weeks 1–2)

Finalise the thesis framing; complete the integrated navigate → dock → perceive → grasp → transport → place → stack cycle into a reliable, repeatable run. Freeze the simulation testbed used for all subsequent experiments. Begin literature review.

Phase 2 — Simulation Experiments (Weeks 2–6)

Execute the systematic trial campaign ($N \geq 20$ per configuration), recording task success rate, placement accuracy, planning time, and cycle time. Conduct the reachability study (effect of mounting, base stop distance, and workspace layout) and the stacking-precision study. Complete literature review (Chapter 2).

Phase 3 — Simulation-to-Reality Transfer (Weeks 5–10, contingent)

Set up the physical LIMO Cobot (onboard computer, drivers, arm and camera calibration), port the pipeline to hardware, and perform the real-robot trials. Compare real-world placement precision against simulation and record the engineering effort required for the transfer. *Contingent on hardware availability and lab access.*

Phase 4 — Analysis (Weeks 9–11)

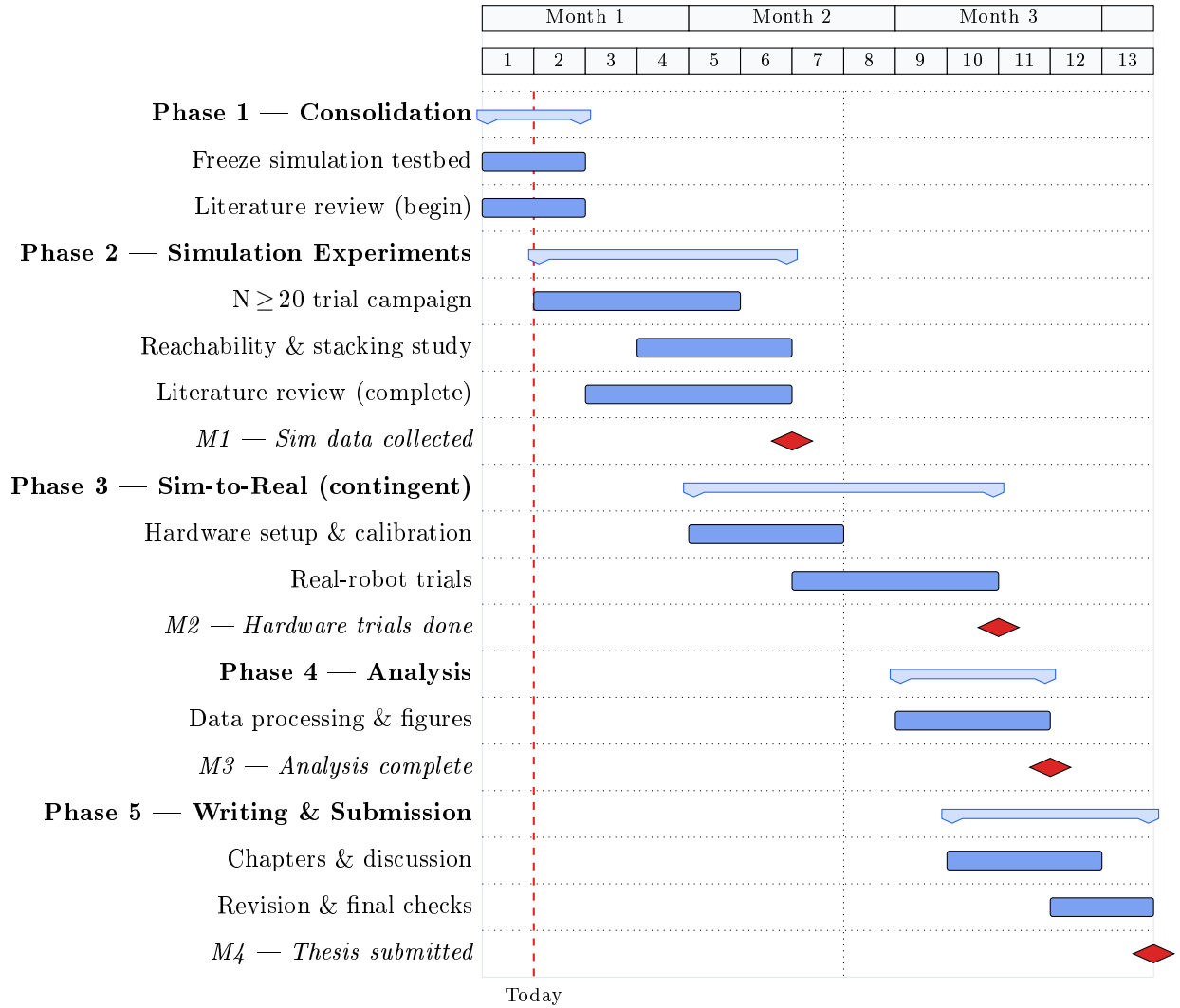
Process and interpret collected data, produce figures and tables, identify dominant failure modes, and prepare results for write-up.

Phase 5 — Dedicated Writing & Submission (Weeks 10–13)

A three-week period reserved for writing and final revision. Complete all chapters, the discussion and conclusion, and the abstract; finalise and submit.

Gantt Chart

Total thesis period: 3 months (13 weeks)



Milestones

Milestone	Timing	Deliverable
M1 — Sim data collected	End of Week 6	$N \geq 20$ simulation runs complete; metrics logged.
M2 — Hardware trials done	End of Week 10	Real-robot stacking trials complete; comparison data logged.
M3 — Analysis complete	End of Week 11	All figures, tables, and failure-mode analysis ready.
M4 — Thesis submitted	End of Week 13	Final corrected thesis submitted.

Overview

Mobile manipulation brings together two abilities that are normally studied apart: moving through an environment and handling the objects in it. A mobile manipulator is a wheeled base with a robotic arm, so one robot can drive to an object, pick it up, carry it, and put it down. The idea is well established and has been demonstrated on many research platforms over the years [?]. Making it work reliably on a small, low-cost robot is still demanding, because the base and the arm have to be coordinated precisely for each task to succeed.

This thesis designs a complete pipeline for such a robot and studies how well it picks objects up, places them, and stacks them. The robot drives to a target, finds it with an onboard camera [?], plans a safe grasp around the surfaces near it, picks the object up, carries it, and places it where it should go. By repeating this cycle and setting each object on top of the last, the robot builds a stack. Stacking is the most demanding goal of the three, because the objects have to be placed precisely enough to rest stably on one another, so it sets the standard the rest of the pipeline has to meet. The work is done first in simulation and then, as far as time and hardware allow, on the real robot. The methods are kept simple and transparent rather than learning-based, so the system is easy to follow and to transfer to hardware. Performance is judged by how often the task succeeds, how accurately objects are placed, and how long planning and each full cycle take. Comparing the simulated behaviour with the behaviour on the real robot is itself one of the things the thesis sets out to measure.

Platform Architecture

The mobile manipulator platform combines an **AgileX LIMO** differential-drive base with an **Elephant Robotics myCobot 280** 6-DoF arm (the LIMO Cobot configuration). The base provides autonomous navigation capabilities, while the arm extends approximately 280 mm with a payload capacity of 250 g. An adaptive parallel-jaw gripper is mounted at the end-effector, and an RGB-D camera provides visual perception. The entire system operates under **ROS 2 Humble** with **Gazebo Classic** as the simulation environment, enabling physics-based testing before hardware deployment. This compact configuration represents a low-cost platform for mobile manipulation research, with the arm’s limited reach creating stringent requirements for base positioning accuracy.

References

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- Macenski, S. et al., “The Marathon 2: A Navigation System,” *IROS*, 2020.
- Masannek, M. et al., “Embodied Autonomous Intelligence for On-Demand Manufacturing,” EAI-WS Proposal, 2026.
- Chaumette, F. and Hutchinson, S., “Visual Servo Control, Part I: Basic Approaches,” *IEEE Robotics & Automation Magazine*, 2006.